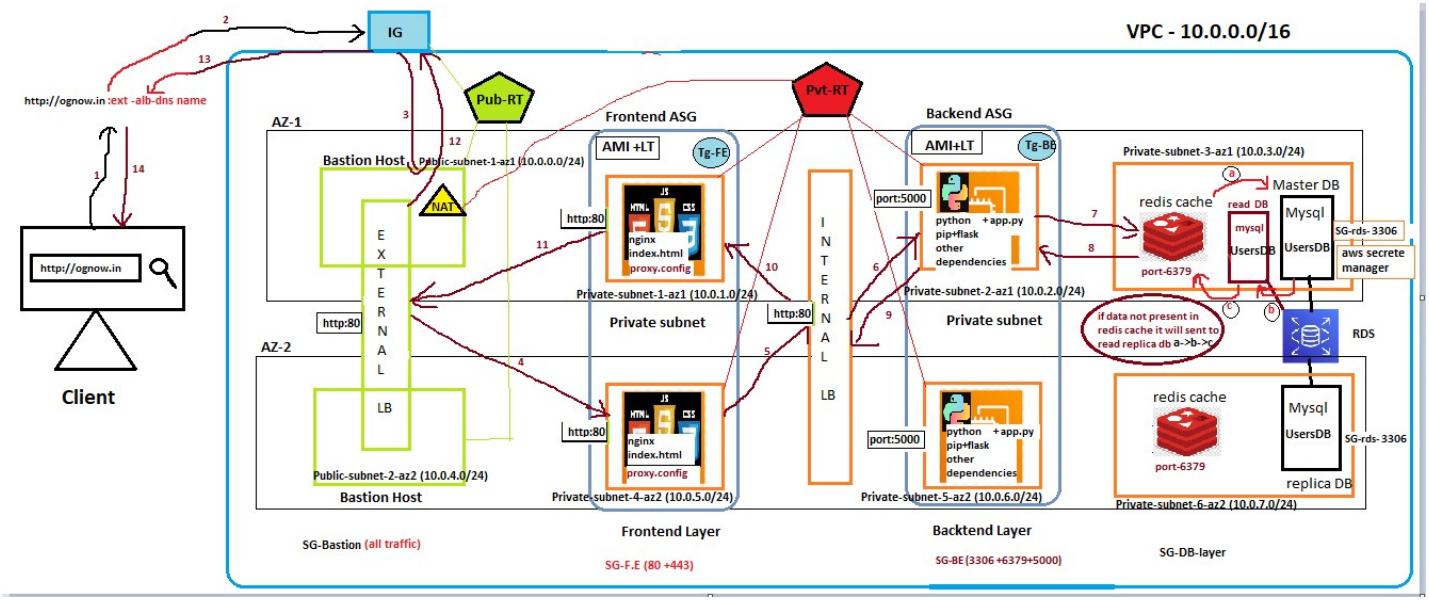
Deploy a 3 tier Fullstack app with python backend + RDS cache



Prerequisite

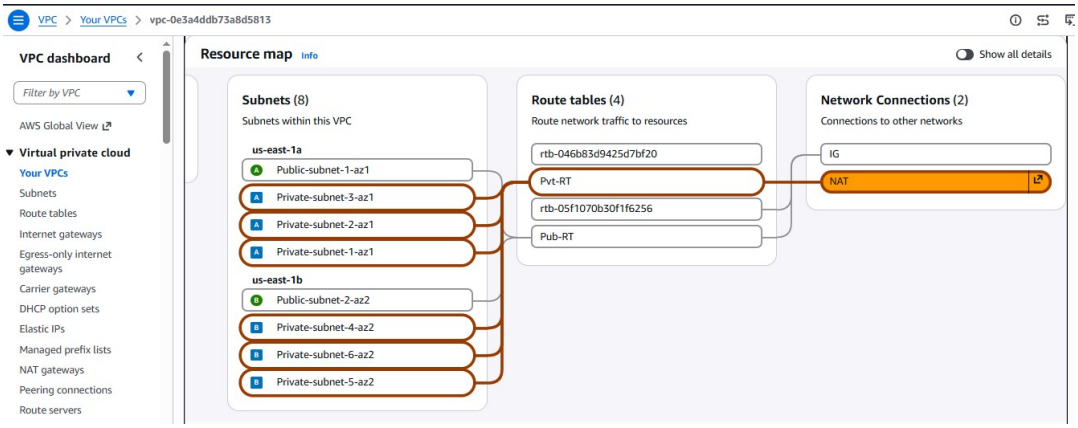
1. VPC (10.0.0.0/16)+IG+NAT
2. 2 availabaility zone (AZ-1 +AZ-2)
3. Create total 8 subents (4 for each az)
   a. Az1 (1 public +3private)
   b. Az2 (1 public +3 private)
4. 2 bastion host + 2 pvt frontend server + 2 pvt backend server)
5. RDS users DB +(1 RDS master in az1 + 1 RDS replica in az2
6. Public-RT + Private-RT
7. 1 FE-ASG (AMI + LT) + 1 BE-ASG (AMI +LT).
8. 1 Internal ALB (BE-Tg1)+ 1 External ALB (FE-Tg1).
9. AWS secrete manager + IAM role
10. Get all required resources from – https://github.com/itmukesh-16/python-backend-testing.git

Az1		Az2	
Public → 10.0.0.0/24	**Public-subnet-1-az1**	**Public → 10.0.4.0/24**	**Public-subnet-1-az2**
Frontend → 10.0.1.0/24	**Private-subnet-1-az1**	**Frontend → 10.0.5.0/24**	**Private-subnet-4-az2**
Backend → 10.0.2.0/24	**Private-subnet-2-az1**	**Backend → 10.0.6.0/24**	**Private-subnet-5-az2**
DB → 10.0.3.0/24	**Private-subnet-3-az1**	**DB → 10.0.7.0/24**	**Private-subnet-6-az2**

Implementation

Infrastructure & Network Set up-

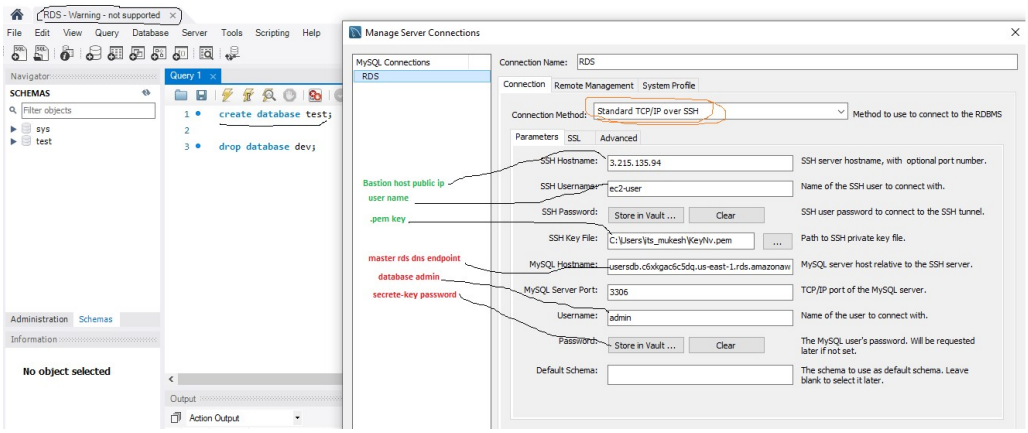
1. Create VPC (10.0.0.0/16) and attach IG and NAT
2. Connect Public-RT to IG and Private-RT to NAT (**Make sure the NAT connected to one of the public subnet for internet supply and while creating select Regional option**)
3. Connect Public-RT →Public subnet , Private-RT → Private subnets.
4. Create 4 SGs, as of now give all access from anywhere for all SGs (SG-Bastion , SG-FE , SG-BE , SG -DB) .letter we can open required port only



[create initially one bastion host , check IG connection and deploy 1 private server , check NAT connection] once success fully tested the connection go to the next phase.

RDS database set up -

5. Create one RDS subnet group .
6. Deploy 1 RDS DB named usersDB in az1 private subnet (Enable aws managed secrete keys)
7. Test connection to RDS DB using mysql workbench via Bastion Host.(mysql workbench →bastion host → RDS DB) .



8. Now create 1 replica usersDB in az1 +1 replica usersDB in az2 (test connection for read DB deployed in both AZs.)
9. Create a redis cache for Masterdb (usersDB).**make sure the SG attached to cache db opens port 6397**

Backend setup –

1. Deploy Backend-az1 server in private-backend-subnet ,check internet connectivity
2. Connect Frontend server in az1 and switch to root user
3. install [python+pip+mariadb105] ex - **#yum install pip -y**
4. Create a directory **backend** -> then create backend.py ,requirements.txt , usersdb.sql .
5. **requirements.txt** {flask + flask_cors + mysql-connector-python + redis + cryptography + pymysql+boto3}
6. under usersdb.sql give users table schema.

python-backend-testing / backend-basic / test.sql

CloudTechDevOps Create test.sql

Code Blame 14 lines (12 loc) · 362 Bytes

```
1  CREATE DATABASE IF NOT EXISTS dev;
2  USE dev;
3
4  CREATE TABLE users (
5      id INT AUTO_INCREMENT PRIMARY KEY,
6      name VARCHAR(100) NOT NULL,
7      email VARCHAR(100) NOT NULL UNIQUE
8  );
9
10 INSERT INTO users (name, email) VALUES
11 ('John Doe', 'john@example.com'),
12 ('Alice Smith', 'alice@example.com'),
13 ('Bob Johnson', 'bob@example.com'),
14 ('Eve Adams', 'eve@example.com');
```

7. Using **mariadb105** (mysql agent) test connection to RDS usersdb .(for private server to RDS usersdb)

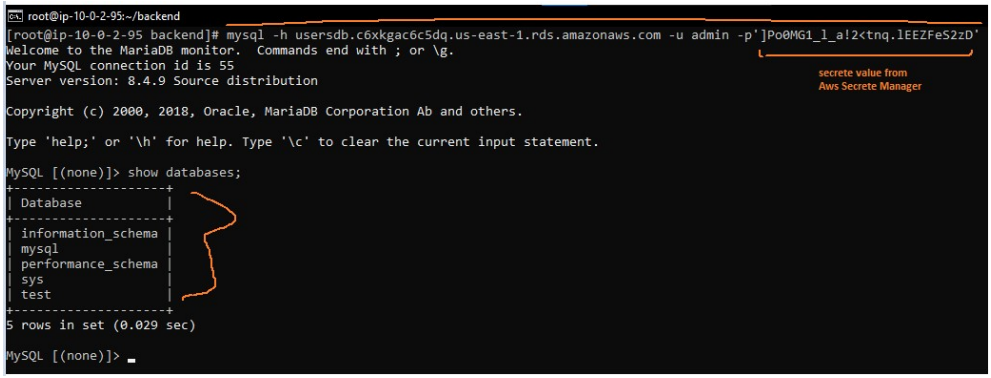
   [at the RDS time, if RDS credentials – self managed]

   To execute sql query by staying in the backend private server
   #mysql -h <dns endpoint> -u admin -p'AdminMukesh'< usersdb.sql

   To manage rds usersdatabase in the backend
   #mysql –h <endpoint dns> -u admin –p

   [*if the RDS creation time , RDS credentials managed by – aws secrete manager , then we need to create an IAM role with permission (***SecretsManagerReadWrite***) and add it to backend server* .otherwise it wont allow to acces]

   - Secrete manager >



8. Include backend code in **backend.py** then run **#python3 backend.py**
9. For temporary fetch the result , open new terminal then check process(ps aux | grep python) and port(ss -tuln) ,whether the python service is running or not if running.
10. then fetch user data from usersdb by –

```
11. curl -X GET http://localhost:5000/users
```

    if there is no out put or you get request : *timed out* ,it means EC2 is not able to connect to the cache database .(check if SG of redis cache allowing SG port: 6379 , subnet , VPC)

    To effieciently handle the process in the same server ,you can configure #pm2 for prcess to run. Also you can create a service and configure for autostart on reboot.(refer - backend service autostart.txt)

12. Then #systemctl enable python3 backend.py

Frontend Set up-

13. Create Frontend server in az1 , connect and switch to root
14. Configure (nginx +index.html +proxy.conf)
15. Install nginx and replace frontend code in **/usr/share/nginx/html/index.html**
16. Configure index.html , make sure the front end code satisfy the structure which will present the data fetched from the backend on the web browser.otherwiser it wont show in browser.
17. Configure proxy.conf file to support reverse –proxy ie redirect the request from external ALB → fronted server → internal ALB. [**reverse-proxy.conf**]

```
server {
    listen 80;
    server_name _;

    # 🔥 Proxy users endpoints directly
    location /users {
        proxy_pass http://internal-ALB-BE-1406133337.us-east-1.elb.amazonaws.com;
        proxy_http_version 1.1;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        proxy_redirect off;
    }
```

Why Reverse-proxy?

when user search the application domain/ip the browser, frontend will deliver the static code to the into browser.
When the browser sent some request to the application ,the request will go the front end server again , and listen at port 80.
Then using reverse-proxy it will send the request to the Internal Load balancer.Later internal ALB will sent that request to healthy target of BE-Tg1:5000 .
Finally the request will reach to backend destination server at port:5000.
if we had not implemented this reverse-proxy , then all request from browser would will try to send request to backend private ip . which is not possible.
But using proxy the front end allows the traffic to the pvt backend server via frontend private server.

18. Now run → #systemctl start nginx
19. Again →#systemctl enable nginx (to support autostart on server reboot)
20. For testing ,try to fetch the data from backend

 # curl -X GET http://backend-server-private-ip:5000/users
If its fetching data it means it backend is working properly

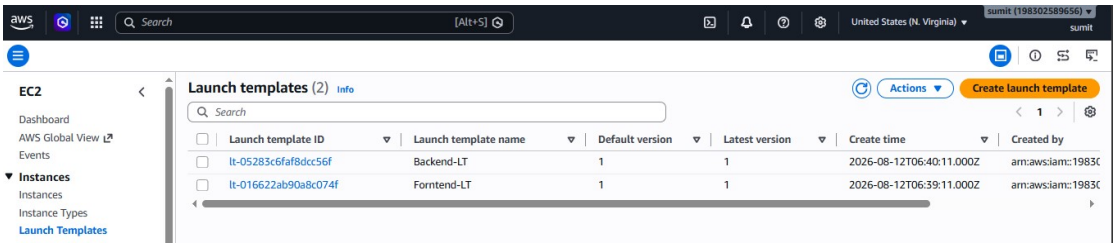
Create AMI and Launch Template –

[Before creating LT , make sure the Frontend and backend services are running ie nginx , python . also make sure you configured the services in such a way it will autostart the services everytime there is a reboot , deploy new instance via ASG,system chrash or new session

For ex nginx #systemctl enable nginx]

[don't create LT directly from running instance , first create AMI ->LT->then attach to ASG]

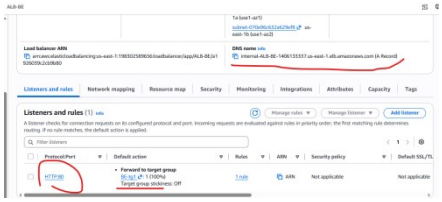
21. Select Frontend-az1 server >AMI>Launch Template (select desired vpc,subnets azs for deployment)>Frontend-LT
22. Using Frontend-LT deploy another frontend server in **Private-subnet-4-az2.**
23. Then do for Backend-az1 server>AMI>Launch Template>Backend-LT
24. Using Backend-LT deploy another frontend server in **Private-subnet-4-az2.**

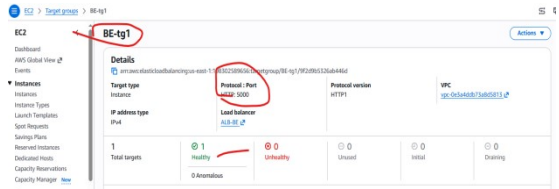


Configure Load Balancer

Internal load balancer for Backend server HA:

a. Create a target group BE-Tg1, port :5000(select the two backend server from both AZs)
b. Give path '/', http protocol and port :5000 (cause python flask running on port 5000) and it will check the port health to distribute the traffic.
c. Now create an application load balancer(ALB) **internal-facing** configure port 5000.
d. Add target group **BE-Tg1** and create internal load balancer **internal-dns**.





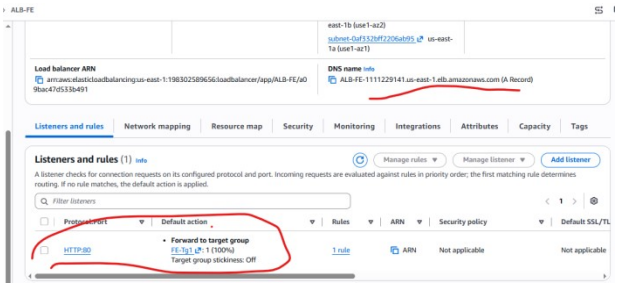
e. Now for test fetch the data using **internal-alb-dns** from frontend server.
 # curl -X GET http://internal-alb-dns/users

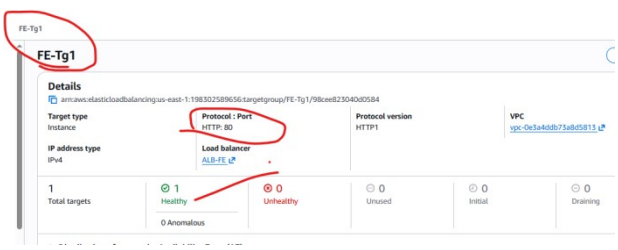
```
}
[ec2-user@ip-10-0-1-243 ~]$ curl -X GET http://internal-ALB-BE-1406133337.us-east-1.elb.amazonaws.com/users
{
  "data": [
    {
      "email": "eve@example.com",
      "id": 4,
      "name": "Eve Adams"
    },
    {
      "email": "john@example.com",
      "id": 5,
      "name": "John Doe"
    },
    {
      "email": "itsmukeshgouda@gmail.com",
      "id": 7,
      "name": "Mukesh Gouda"
    }
```

- **If the test successful.then update the internal-alb-dns link in revers_proxy.conf file of the frontends server.or you can create new Launch Template for backend.**
- **Otherwise the frontend will always point to fixed private backend server ip ,as a result the load balancer will not work properly**
- **Also update Frontend-LT or else it will contain the old server configuration and will cause issue in ASG .**

External/Internet load balancer for Frontend server HA:

a. Create a target group FE-Tg1(select the two backend server from both AZs)
b. Give path '/', http protocol and port :80 (cause nginx running on port 80) and it will check the port health to distribute the traffic.
c. Now create an application load balancer(ALB) **External-facing** configure port 80.
d. Add target group **FE-Tg1** and create External load balancer **Ext-alb**.
e. Now access the **External-alb-dns** in browser and see if the the index page is properly working or not.





Now create ASG-

[make sure the Launch Template you providing to ASG has latest/updated configuration of running instances ,otherwise it will not deploy the updated instance configuration and services running inside the servers . As a result , it might not give expected result]

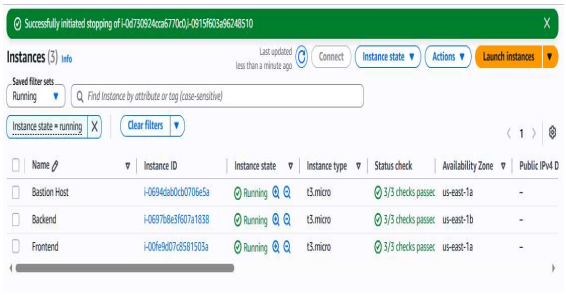
- Configure Frontend -ASG for Frontend server HA

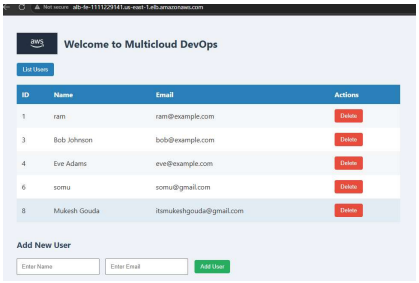
- Configure Backend – ASG for backend server HA

[don't delete the source instance from which Instance template created ,once the ASG created you can delete the source instance if required]

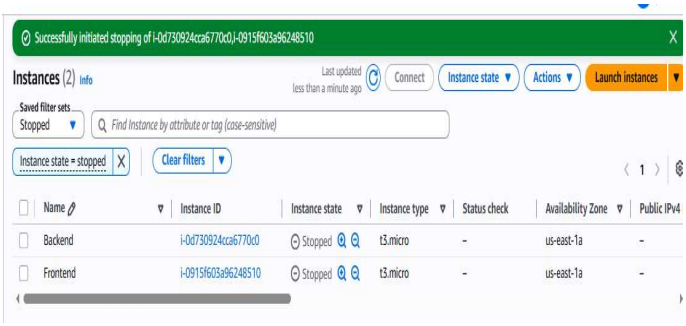
For testing purpose lower the cpu utilisation threshold or stop the source instance and see if the autoscalling is working or not.

Source instances running

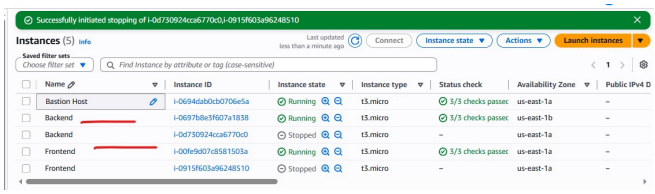




Now stop the source isntances



Observer the Autoscalling



Now configure SGs-

Since we had opened all the ports for all the sercurity groups for testing and its not recommended .it may cause security breach .

Hence only open required inbound and outbound ports for all the SGs

Final Security Groups

#	Security Group	Attach to	Required Inbound	Required Outbound
1	`SG-Bastion`	Bastion Host	`22` ← Your IP	`22` → `SG-FE`, `SG-BE`
2	`SG-ALB-FE`	Internet-facing ALB	`80,443` ← Internet	`80` → `SG-FE`
3	`SG-FE`	Frontend ASG	`80` ← `SG-ALB-FE` `22` ← `SG-Bastion`	`80` → `SG-Internal-ALB` `443` → Internet
4	`SG-Internal-ALB`	Internal ALB	`80` ← `SG-FE`	`5000` → `SG-BE`

5	`SG-BE`	Backend ASG	`5000` ← `SG-Internal-ALB`
			`22` ← `SG-Bastion`
				`3306` → `SG-RDS`
				`6379` → `SG-Redis`
				`443` → Internet

6	`SG-Redis`	ElastiCache Redis	`6379` ← `SG-BE`	No application outbound required
7	`SG-RDS`	RDS MySQL	`3306` ← `SG-BE`	No application outbound required

Then again try to access the application

Task Over